

Concerns about artificial intelligence feedback validation in the 2-Sigma platform

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To the Editor,

I read with great interest the report by Turner et al. describing the 2-Sigma artificial intelligence (AI)-based simulation platform for personalized medical education.¹ However, I am concerned that the article does not address the verification of the quality of AI-generated feedback, which is a critical flaw.

The authors built 2-Sigma using chain-of-thought prompting and few-shot learning with OpenAI's GPT-4 API, providing immediate feedback on clinical reasoning across 1603 sessions.^{1,2} While these technical approaches are promising, I note that research on large language models in education emphasizes that such complexity increases risks of misinterpreting student reasoning. Since there is currently no publicly available verification data comparing the evaluation results of AI with those of expert manual reviews, I am concerned that educators will not be able to determine whether this feedback can accurately identify reasoning errors or merely matches students' answers against preset patterns.³

The observed correlation between the increase in questions and error diagnosis requires careful interpretation, particularly for viral myocarditis and hypersensitivity pneumonitis cases. This pattern may reflect a lack of knowledge, but I suspect it may also indicate that the performance of artificial intelligence lacks clinical authenticity or that feedback fails to guide effective exploration. Recent literature has warned that automated systems may struggle to distinguish between reasonable alternative reasoning and genuine errors, which could reinforce incorrect decision-making patterns.⁴

The article describes red team testing focused on safety, but does not elaborate on teachers' review of the accuracy of AI evaluations. I question how many conversation records have been verified by experts and what processing procedures exist for non-standard terminology. Before scaling AI tutors, I believe our community needs rigorous validation studies demonstrating that feedback is educationally sound and clinically accurate.

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Conflicts of interest

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Disclaimers

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Data availability

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Letter to the Editor